

Q.1

$$x_1 \leq 2 - x_2$$

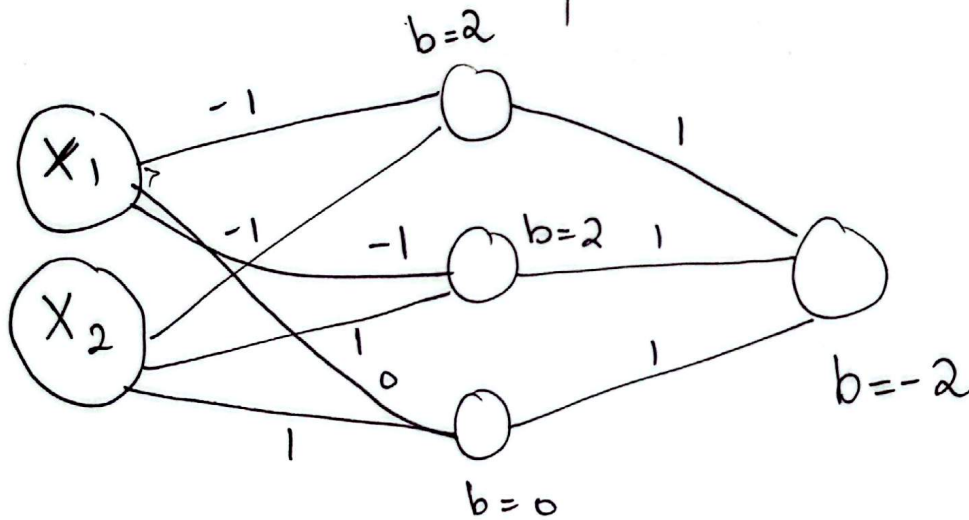
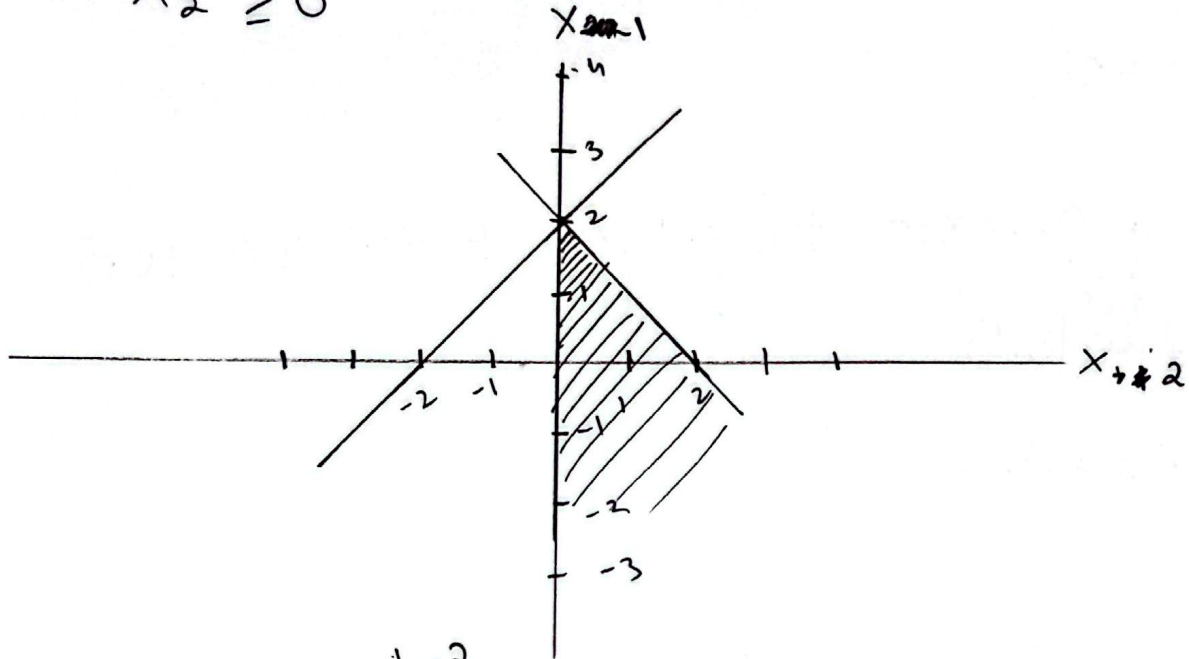
$$x_1 \leq x_2 + 2$$

$$x_2 \geq 0$$

$$x_1 + x_2 \leq 2 = \frac{x_1}{2} + \frac{x_2}{2} \leq 1$$

$$x_2 - x_1 \geq -2 =$$

$$x_1 - x_2 \leq 2 \rightarrow \frac{x_1}{2} - \frac{x_2}{2} \leq 1$$



Bipolar Activation at output
and hidden layer

Q.2

$$Z_1 = 0.1 \times 10 + 0.3 \times 40 + -0.5 \times 50 + 0.5$$

$$Z_1 = -11.5$$

$$h_1 = 0$$

$$Z_2 = 0.2 \times 10 + (-0.4 \times 40) + (0.5 \times 50) + 0.5$$

$$Z_2 = 16.5$$

$$h_2 = 16.5$$

$$y_1 = 0.7 \times 0 + 0.9 \times 16.5 + 0.5$$

$$y_1 = 15.35$$

$$O_1 = \frac{1}{1 + e^{-15.35}} = 0.999$$

$$y_2 = 0.8 \times 0 + (0.1 \times 16.5) + 0.5$$

$$y_2 = 2.15$$

$$O_2 = 0.895$$

$$\Delta w_7 = (0.1) [(0.99)(0.99)(1-0.99) \times 0] = 0$$

$$w_7 = 0.7$$

$$\Delta w_8 = 0$$

$$w_8 = 0.8$$

$$\Delta w_9 = (0.1) [(0-0.99) 0.99 (1-0.99) (16.5)]$$

$$\Delta w_9 = -0.016$$

$$w_9 = 0.9 - 0.016$$

$$w_9 = 0.883$$

$$\Delta w_{10} = (0.1) [(1-0.89)(0.89)(1-0.89) 16.5]$$

$$= 0.017$$

$$w_{10} = 0.1 + 0.017$$

$$w_{10} = 0.117$$

$$\Delta b_1^2 = -9.801 \times 10^{-4}$$

$$b_1^2 = 0.5 - 9.8 \times 10^{-4} = 0.499$$

$$b_1^2 = 0.499$$

$$\Delta b_2^2 = 1.07 \times 10^{-3}$$

$$b_2 = 0.501$$

$$\Delta w_1 = 0$$

$$w_1 = 0.1$$

$$\Delta w_3 = 0$$

$$\Delta w_5 = 0$$

$$\Delta b_1' = 0$$

$$w_3 = 0.3$$

$$w_5 = 0.5$$

$$b_1' = 0.5$$

$$\begin{aligned}\Delta w_2 &= 0.1 \left[-8.82 \times 10^{-3} + 1.0769 \times 10^{-3} \right] \times 10 \\ &= -7.743 \times 10^{-3}\end{aligned}$$

$$w_2 = 0.192$$

$$\Delta w_4 = -0.030$$

$$w_4 = -0.430$$

$$\Delta w_6 = -0.038$$

$$w_6 = 0.561$$

$$\Delta b_2' = -7.743 \times 10^{-4}$$

$$b_2' = 0.499$$

$$\Delta w_7 = (0.1) [(0-0.99)(0.99)(1-0.99) \times 0] = 0$$

$$w_7 = 0.7$$

$$\Delta w_8 = 0$$

$$w_8 = 0.8$$

$$\Delta w_9 = (0.1) [(0-0.99)(0.99)(1-0.99)(16.5)]$$

$$\Delta w_9 = -0.016$$

$$w_9 = 0.9 - 0.016$$

$$w_9 = 0.883$$

$$\Delta w_{10} = (0.1) [(1-0.89)(0.89)(1-0.89) 16.5]$$

$$= 0.017$$

$$w_{10} = 0.1 + 0.017$$

$$w_{10} = 0.117$$

$$\Delta b_1^2 = -9.801 \times 10^{-4}$$

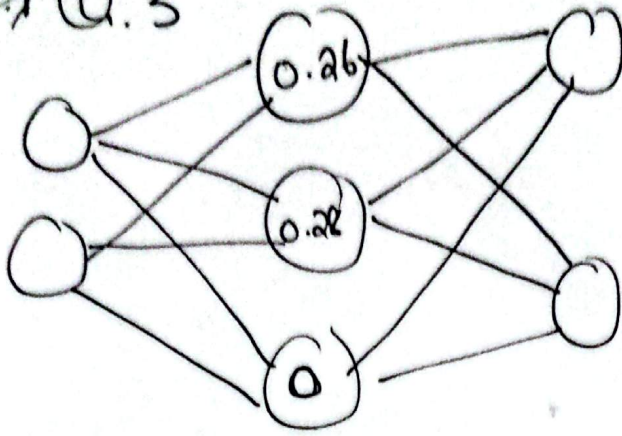
$$b_1^2 = 0.5 - 9.8 \times 10^{-4} = 0.499$$

$$b_1^2 = 0.499$$

$$\Delta b_2^2 = 1.07 \times 10^{-3}$$

$$b_2 = 0.501$$

Q.1 Q.3



$$(0.5)(0.2) + (-0.2 \times -0.3) + 0.1$$

$$= 0.26$$

$$(0.5)(0.4) + (-0.2)(0.1) + 0.1$$

$$= 0.28$$

$$(0.5)(-0.5) + (-0.2)(0.2) + 0.1$$

$$= -0.19 \approx 0$$

$$(0.26)(0.5) + (0.28)(-0.2) + (0.0.1) + 0.2$$

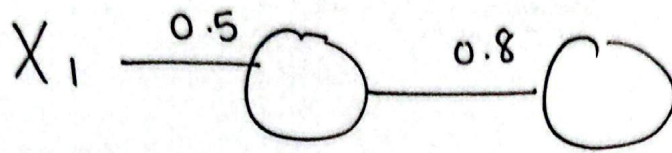
$$= 0.274 = \frac{1}{1+e^{-0.274}} = 0.568 = y_1$$

$$(0.26)(0.1) + (0.28 \times 0.3) + 0.2$$

$$= 0.31$$

$$\frac{1}{1+e^{-0.31}} = 0.57688 = y_2$$

Q.4



$$1(0.5) = 0.5$$

$$\frac{1}{1+e^{-0.5}} = 0.6224$$

$$0.6224(0.8) = 0.4979$$

$$\frac{1}{1+e^{-0.4979}} = 0.6219$$

$$1) E = \frac{1}{2} (0.5 - 0.6219)^2 = 0.007429$$

$$2) \delta = (0.6219)(1 - 0.6219)(0.5 - 0.6219) = -0.028663$$

$$\Delta w_2 = (0.1)(0.6224)(-0.028663) = -0.0017839$$

$$\text{new } w_2 = -0.0017839 + 0.8 = 0.7982$$

$$3) \delta = (0.6224)(1 - 0.6224)(0.8x - 0.028663)$$

$$\delta = -0.005389$$

$$\Delta w_1 = (0.1)(-0.005389)(1) = -5.389 \times 10^{-4}$$

$$\text{new } w_1 = 0.5 + (-5.389 \times 10^{-4}) = 0.49946$$